11-W1　速度选择后的荷质比

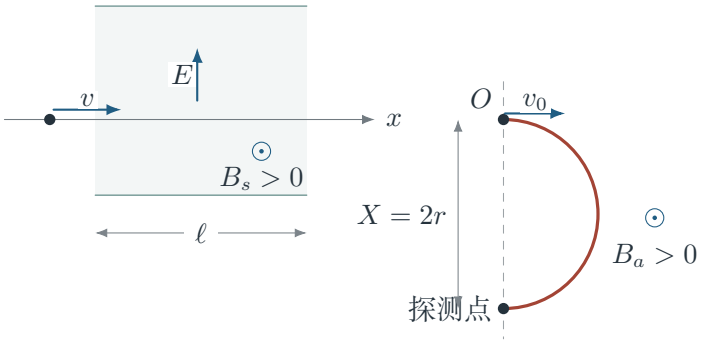


图 11-13　题图　速度选择与磁分析区分开，分析图按正电荷画半圆

$$v_0 = E/B_s$$

$$|q|/m = v_0/(B_a r)$$

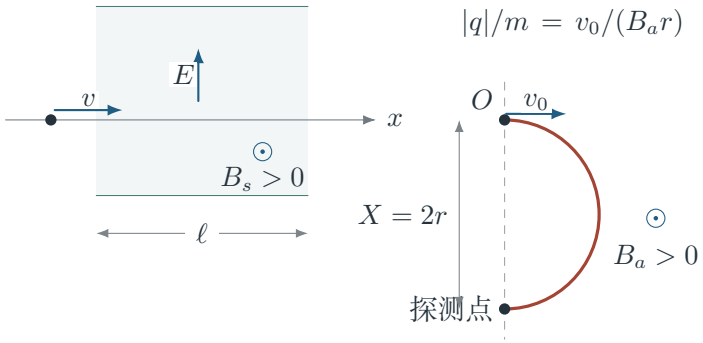


图 11-14　解答分析图　零偏转确定速度，弯转方向判断电荷符号

11-W2　从静止释放的正交场轨迹

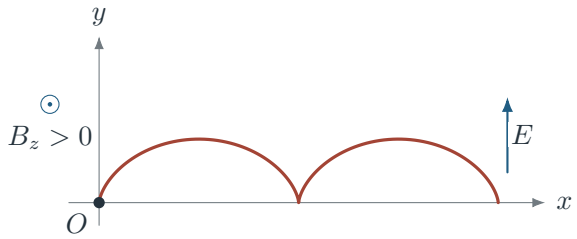


图 11-15　题图　从静止释放，电场向上，磁场向纸外

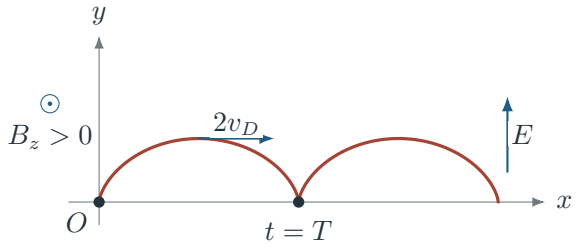


图 11-16　解答分析图　每个整周期速度归零，导引中心仍向右推进

11-W3　点源束流的整周期会聚

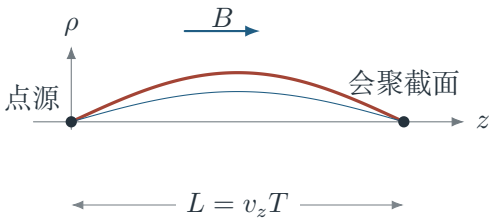


图 11-17　题图　两束粒子的真实离轴距离，半周期时离轴最远，整周期回轴

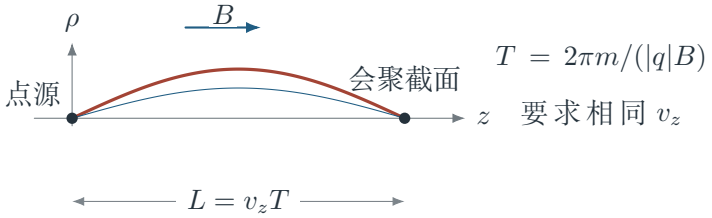


图 11-18　解答分析图　纵向速度决定会聚截面，横向初速决定最大离轴距离

11-W4　回旋加速器的能量与过缝次数

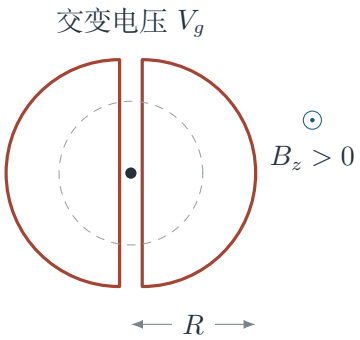


图 11-19　题图　D 形电极内回旋，狭缝提供加速电压

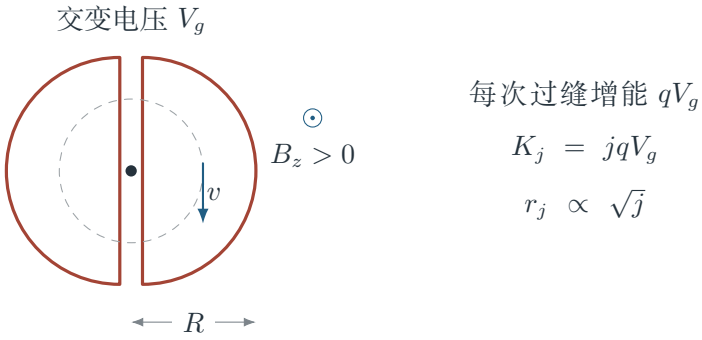


图 11-20　解答分析图　过缝次数与完整回转次数相差约一倍

11-W5　抛物型磁阱的往返运动

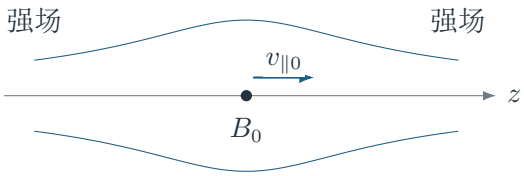


图 11-21　题图　中面场最弱，两端场逐渐增强

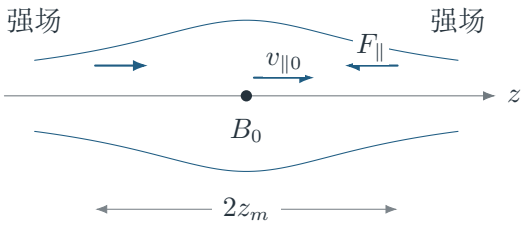


图 11-22　解答分析图　纵向镜力为回复力，镜点对应纵向速度为零

11-W6　环向场的梯度与曲率漂移

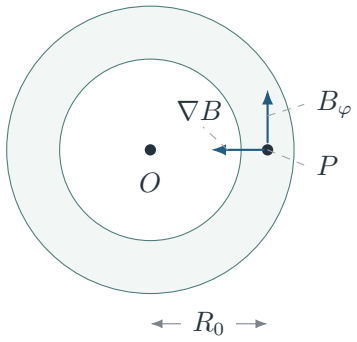
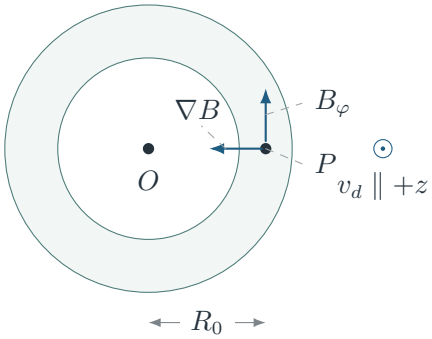


图 11-23　题图　环向场右侧的局部场向上、场强向内增大



分析示意取 $q > 0$

梯度与曲率漂移同向

图 11-24　解答分析图　正电荷漂移向纸外，负电荷漂移反向

11-W7　横向开路与短路的磁电阻

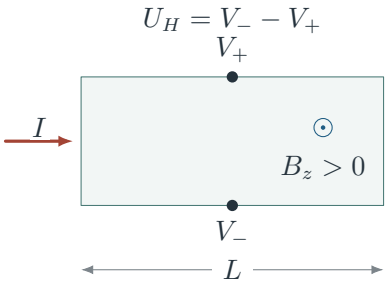
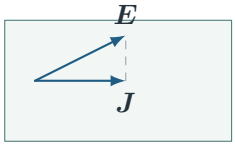


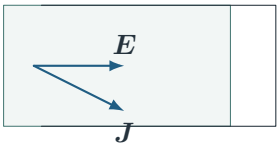
图 11-25　题图　磁场向纸外，横向接线决定开路或短路条件

横向开路：$J_y = 0$



$E_x/J_x = \rho_0$

横向短路：$E_y = 0$



$E_x/J_x = \rho_0(1+\beta^2)$

箭头取 $\beta = q\tau B/m > 0$；B 沿纸外

图 11-26　解答分析图　两种横向接线分别建立霍尔场或允许横向电流，图示正载流子

11-W8　霍尔信号相消时的纵向磁阻

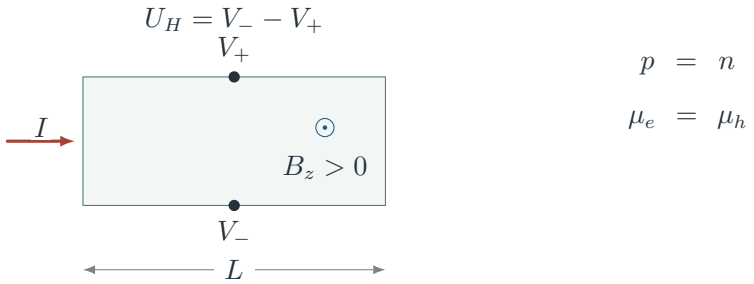


图 11-27　题图　电子与空穴同时导电，浓度和迁移率相等

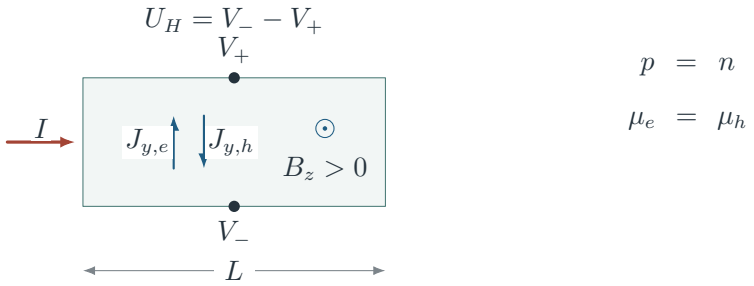


图 11-28　解答分析图　两种侧向电流相消，纵向电导仍受磁场影响

11-W9　均匀磁场的两种矢势

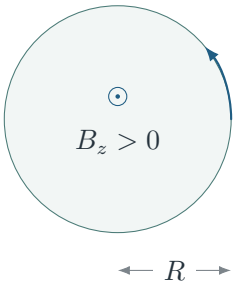
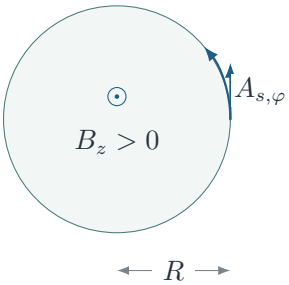


图 11-29　题图　闭合圆周从纸外看逆时针，磁场向纸外



$$A_L = A_s + \nabla\chi$$
$$\chi = Bxy/2$$
$$\oint A \cdot \mathrm{d}l = B\pi R^2$$

图 11-30　解答分析图　规范不同，圆周积分均给出相同磁通

11-W10　反向平行电流的对数矢势

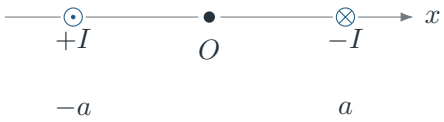
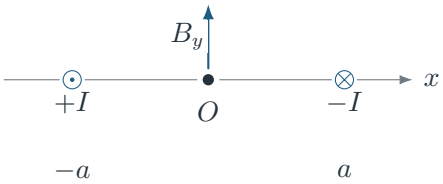


图 11-31　题图　左线电流向纸外，右线向纸内



$$A_z(0,y) = 0$$

$$B_y(0,y) > 0$$

图 11-32　解答分析图　中垂面势值为零，其法向梯度仍非零

11-W11　均匀圆柱体电流的磁矢势

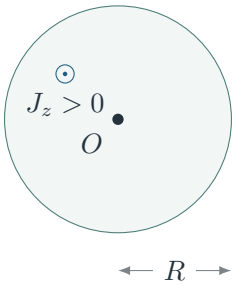
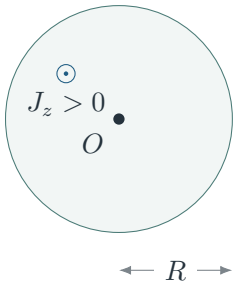


图 11-33　题图　均匀体电流圆柱的横截面



$A_z(R) = 0$

内部二次式

外部对数式

图 11-34　解答分析图　积分 $A_z' = -B_\varphi$，表面势与导数连续

11-W12　由量子霍尔平台求面密度

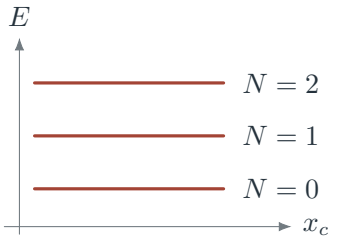


图 11-35　题图　均匀场使二维电子轨道能量等间隔量子化

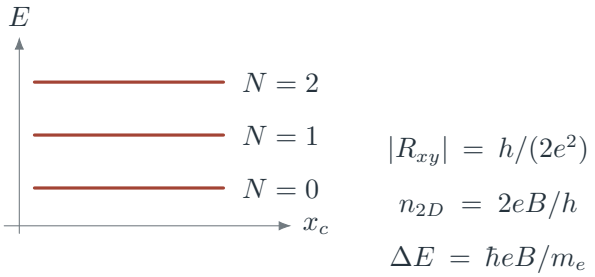


图 11-36　解答分析图　平台读数给面密度，轨道能级间隔依赖质量

11-T1　有限长度选择器的出口偏差

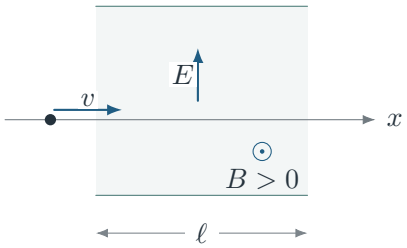


图 11-37　题图　本题只在出口设置狭缝，不限制中途偏离

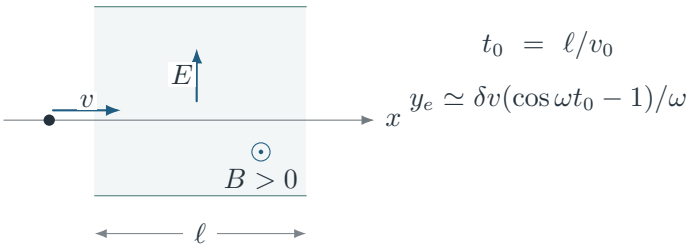


图 11-55　解答分析图　出口偏移依赖飞行时间所对应的回旋相位

11-T2　漂移叠加回旋的轨迹形状

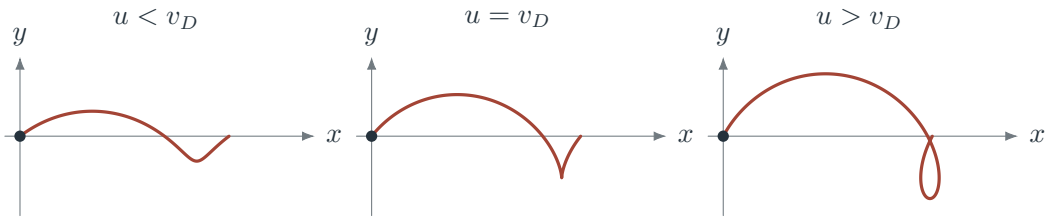


图 11-38　题图　并列同一漂移方向下的三种初速示意，轨迹均由参数方程绘制

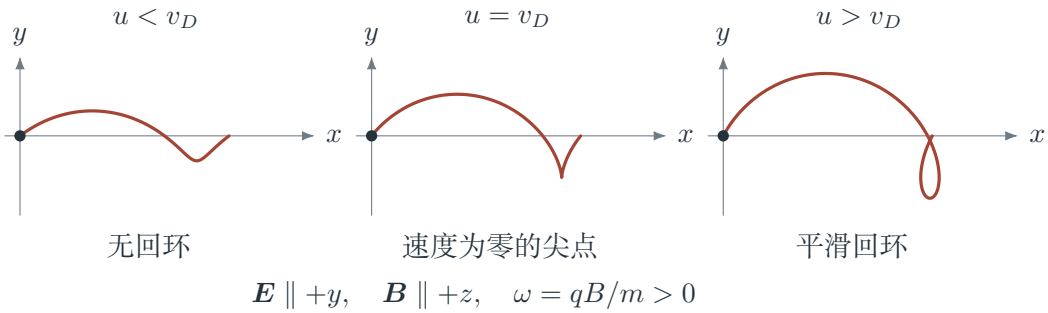


图 11-56　解答分析图　分别检验速度为零和横向回退，尖点与回环对应不同条件

11-T3　磁聚焦的角度容差

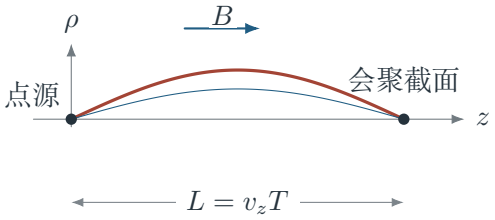


图 11-39　题图　图示相同纵向速度下的离轴距离，角度色散另行计算

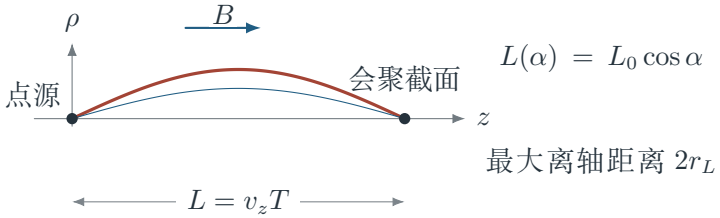


图 11-57　解答分析图　整周期轴向误差与最大离轴距离分别限制束流

11-T4　回旋加速器的累计相位误差

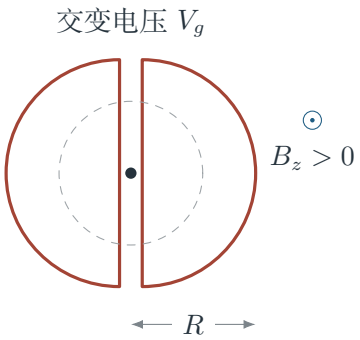
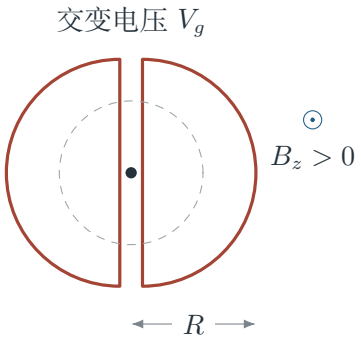


图 11-40　题图　固定射频在每个半圆间隔切换加速方向



$$\delta\varphi_j = \pi K_j/(mc^2)$$

$$\Delta\varphi_N = \sum_j \delta\varphi_j$$

图 11-58　解答分析图　相对论导致到缝迟到，各段相位误差逐次累加

11-T5　含静电势的轴向磁阱

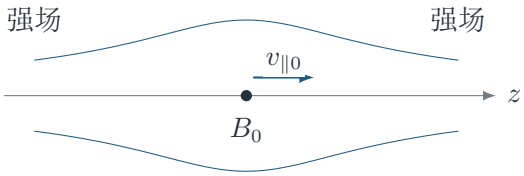
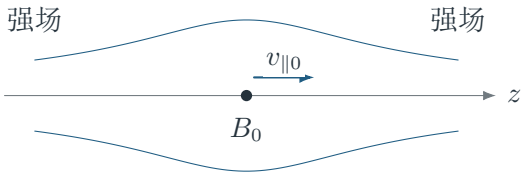


图 11-41　题图　磁镜外还存在沿轴线变化的静电势



$$U_{\mathrm{eff}} = \mu B + q\phi$$

比较 $\mu B_0 + q\Phi$ 与零

图 11-59　解答分析图　电荷符号决定电势增强或削弱纵向约束

11-T6　重力漂移与支承电流

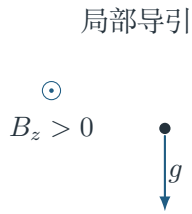
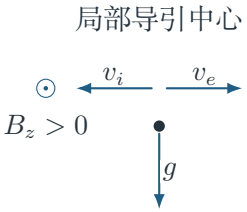


图 11-42　题图　重力向下，磁场向纸外



$J_x < 0$

$J \times B$ 向上

图 11-60　解答分析图　粒子漂移方向相反，约定电流贡献同向

11-T7　损失锥中的粒子数与通量

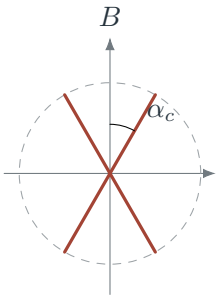
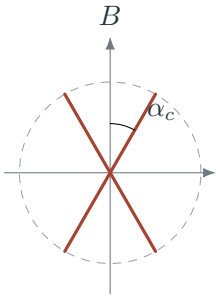


图 11-43　题图　速度方向球面上的两个损失锥，图中为剖面



$$f_N = 1 - \cos\alpha_c$$

$$f_\Gamma = \sin^2\alpha_c$$

图 11-61　解答分析图　计数用立体角，轴向通量另乘 $v\cos\alpha$

11-T8　不等迁移率的补偿半导体

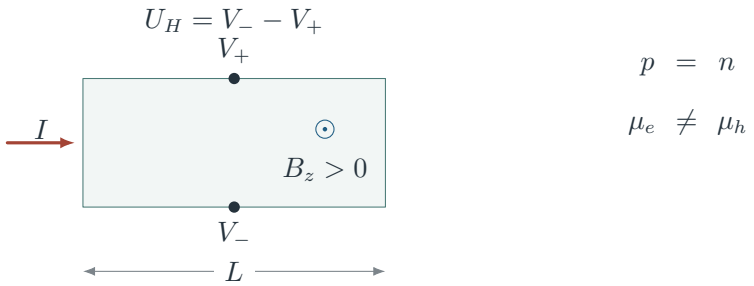


图 11-44　题图　补偿浓度不等于迁移率相同

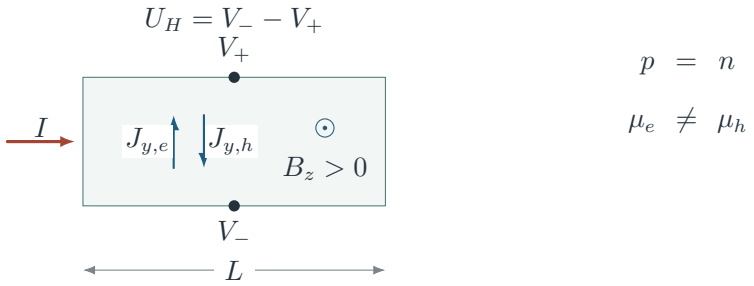


图 11-62　解答分析图　侧向响应权重不同，霍尔信号一般不相消

11-T9　两自旋通道的多层电阻

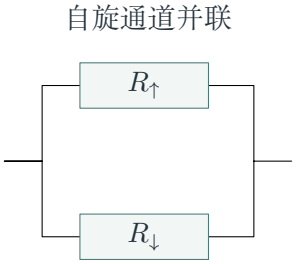


图 11-45　题图　自旋保持时，两通道各自串联后再并联

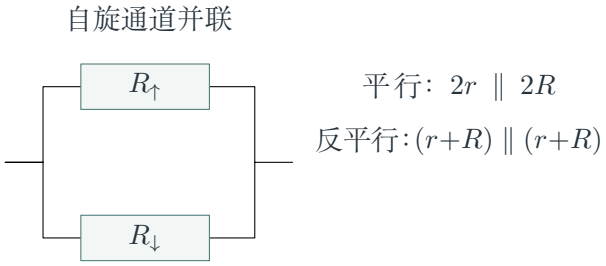


图 11-63　解答分析图　反平行磁化使两个通道都经过一段高阻层

11-T10　开口势积分与闭合磁通

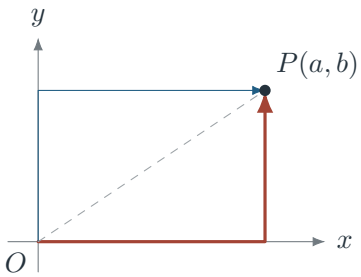
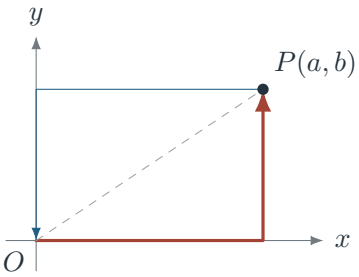


图 11-46　题图　从原点到同一终点的三种路径



$$\Delta\chi = Bab/2$$

闭合路径围住面积 ab

图 11-64　解答分析图　沿红色路径去、反向沿蓝色路径回，法向为纸外

11-T11　无限螺线管外的矢势环流

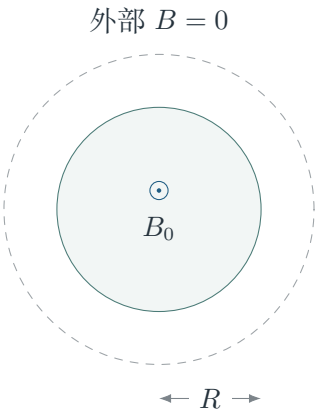
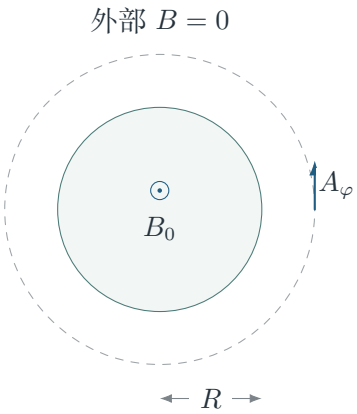


图 11-47　题图　外部圆环围住螺线管，局部磁场为零

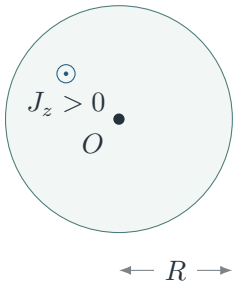


外部 $\nabla \times A = 0$

绕轴 $\oint A \cdot \mathrm{d}l = \pi B_0 R^2$

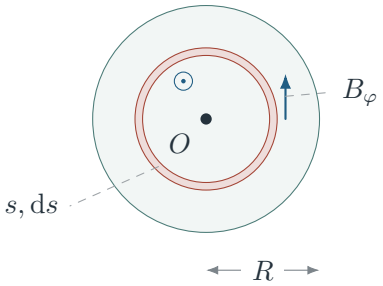
图 11-65　解答分析图　非零闭合积分限制整体单值规范变换

11-T12　非均匀体电流的四次矢势



$$J_z = J_0(1 - r^2/R^2)$$

图 11-48　题图　电流密度从轴心向表面下降



$$J_z = J_0(1 - r^2/R^2)$$

$$J_z > 0$$

图 11-66　解答分析图　累计电流给场，再积分得到四次内部矢势

11-T13　边缘场中的正则角动量

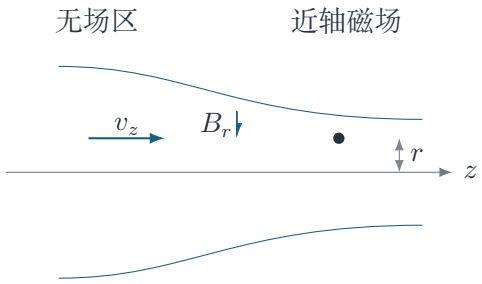
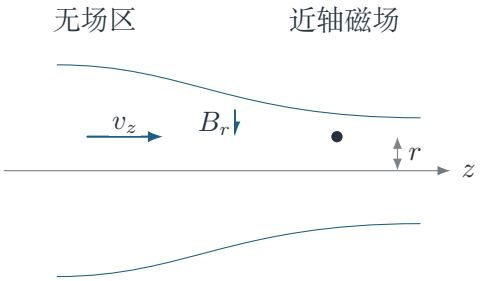


图 11-49　题图　入口轴向增强时，径向场朝轴线收拢



$$P_\varphi = mrv_\varphi + qrA_\varphi$$

$$v_\varphi = -qBr/(2m)$$

图 11-67　解答分析图　入口径向场提供方位力矩，正则角动量守恒

11-T14　磁偶极子的圆周势积分

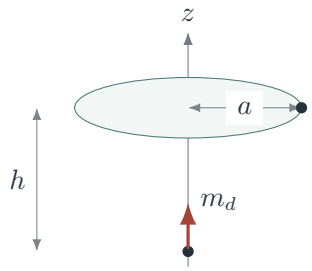
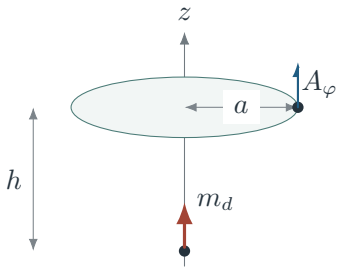


图 11-50　题图　圆周位于偶极子上方，积分圆盘不含源点



$h > 0$：圆盘无奇点

$$\Phi = \oint A \cdot \mathrm{d}l$$

图 11-68　解答分析图　圆周势积分和圆盘磁通可直接相互核对

11-T15　电流微元与完整回路的散度

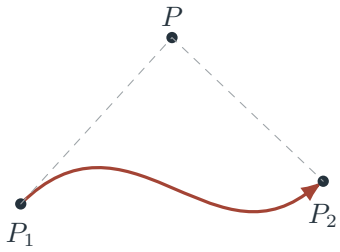
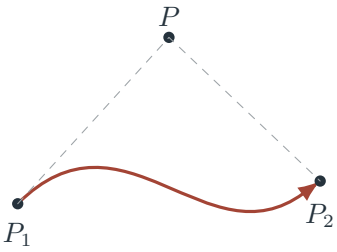


图 11-51　题图　场点到电流段两端的距离不同



开口：保留两端差

闭合：端点项相消

图 11-69　解答分析图　源点全微分揭示单段与完整闭合回路的差别

11-T16　整数填充的磁场与能级尺度

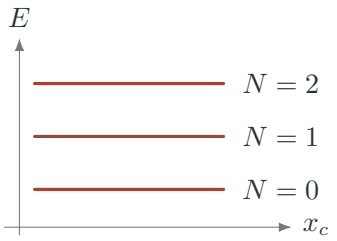


图 11-52　题图　每个自旋支的轨道态数正比于磁场

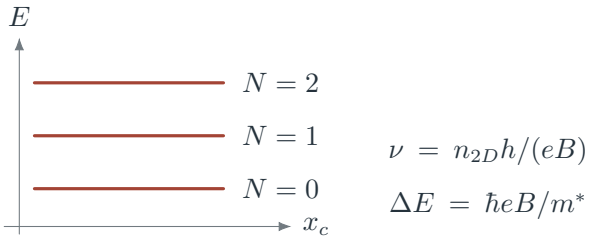


图 11-70　解答分析图　填充条件给磁场，平台宽度还需输运与局域化信息

11-T17　速度选择与半圆质谱（限时综合）

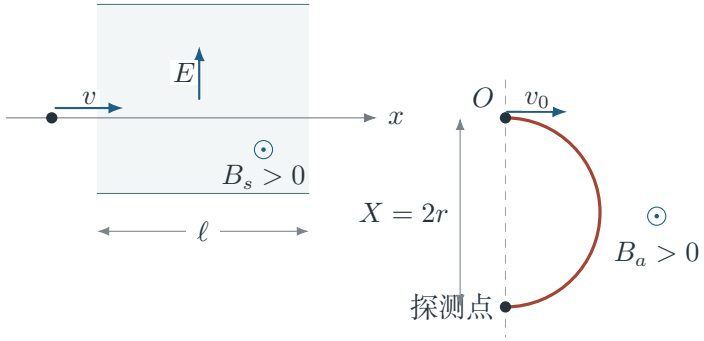


图 11-53　题图　正离子经半圆落在入口下方，落点距离为直径

$$m = eB_aX/(2v_0)$$

$$U = B_aXv_0/4$$

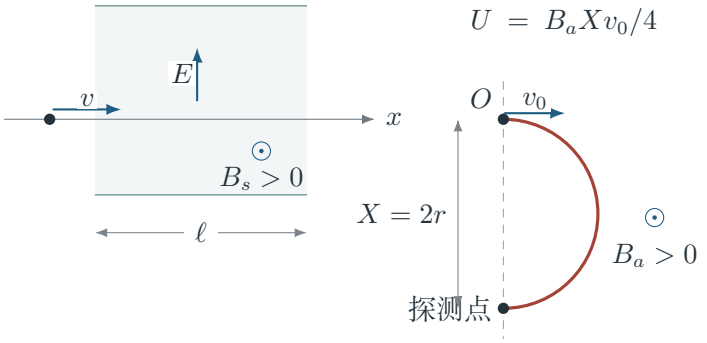
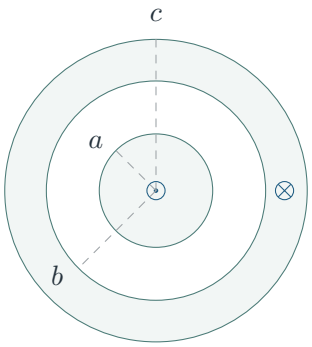


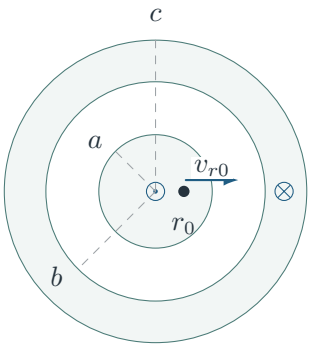
图 11-71　解答分析图　速度选择、加速电压和分析半径必须相互一致

11-T18　同轴电流的矢势与粒子运动（限时综合）



$+I$ 内柱，$-I$ 外壳

图 11-54　题图　内柱电流向纸外，外壳返回电流向纸内



$$A_z(c) = 0$$

$$P_z = mv_z + qA_z$$

K 守恒

$+I$ 内柱，$-I$ 外壳

图 11-72　解答分析图　轴向正则动量守恒，径向能量决定边界可达性